

Active Directory Lab – Final Summary Report

Objective

The objective of this project was to build a small Active Directory lab using Windows Server 2022 and Windows 11. The lab was used to practise common domain administration tasks, including server configuration, domain controller setup, user and group management, Group Policy configuration, domain client joining, and account lockout investigation.

Lab Environment

The lab was built in VirtualBox using a Windows Server 2022 virtual machine and a Windows 11 client virtual machine. The Windows Server machine was configured as the domain controller for the adlab.local domain.

The server used a static host-only IP address of 192.168.56.10, which allowed the Windows 11 client to communicate with the domain controller inside the isolated lab network. The server was also configured with NAT connectivity during setup.

Domain name: adlab.local

NetBIOS name: ADLAB

Tools Used:

- VirtualBox
- Windows Server 2022 Evaluation
- Windows 11
- Server Manager
- Active Directory Users and Computers
- Group Policy Management
- Group Policy Management Editor
- Event Viewer
- Command Prompt

Project Summary

The project began by installing and configuring a Windows Server 2022 virtual machine. The server was configured with two network adapters: one NAT adapter for setup connectivity and one host-only adapter for the internal lab network. A static IP address was assigned to the host-only adapter so that the server could later act as a reliable domain controller and DNS server.

Active Directory Domain Services was then installed through Server Manager. The server was promoted to a domain controller, and a new domain called adlab.local was created. DNS was also configured as part of the domain controller setup so that domain clients could locate and join the domain.

After the domain was created, Active Directory Users and Computers was used to create organisational units, domain users, and a security group. The lab included OUs such as Lab Users, Lab Computers, IT Support, Finance and HR. Test users were created and the Helpdesk User account was added to the IT-Support-Staff security group. This was verified using both Active Directory Users and Computers and Command Prompt.

Group Policy was then configured to apply basic domain security settings. A login banner policy was created and applied, and account lockout settings were configured with a threshold of 5 invalid logon attempts, a 30-minute lockout duration, and a 30-minute reset counter. The gpresult /r command was used to confirm that the Group Policy Objects had applied.

Finally, a Windows 11 client machine was joined to the adlab.local domain. The client appeared as a computer object in Active Directory, and the Helpdesk User domain account was used to log in successfully. The whoami command confirmed that the session was running as adlab\helpdesk.user.

An account lockout test was also performed using the jsmith account. After repeated failed login attempts, the account was locked out. The lockout was confirmed in Active Directory Users and Computers, and Event Viewer recorded the related Windows Security events, including Event ID 4625 for failed logon and Event ID 4740 for account lockout.

Key Findings

The Windows Server virtual machine was successfully prepared as the foundation for the Active Directory environment. The static IP address on the host-only adapter allowed the server to act as a reliable domain controller and DNS server for the lab.

The adlab.local domain was created successfully, and Active Directory Users and Computers confirmed that the domain could be managed through the graphical interface. Organisational units, users, and groups were created successfully. The lab showed how users can be organised into OUs and assigned to security groups for easier administration.

Group Policy was successfully used to apply a login banner and account lockout settings. The login banner appeared before user sign-in, and the account lockout policy was later tested successfully.

The Windows 11 client successfully joined the domain and authenticated using a domain user account. Event Viewer also captured failed logon and account lockout events, showing how Windows Security logs can be used for basic investigation.

Skills Demonstrated

- Windows Server and Active Directory lab setup
- VirtualBox networking, including NAT and host-only adapters
- Static IP and DNS configuration for a domain environment
- Active Directory Domain Services installation and domain controller promotion
- Active Directory user, group, and organisational unit management
- Group Policy configuration and verification
- Windows 11 domain joining and domain user authentication
- Account lockout testing and troubleshooting
- Windows Security log review using Event Viewer
- Investigation of failed logon and account lockout events

Conclusion

This Active Directory lab demonstrated the full process of building and testing a basic Windows domain environment. The project covered Windows Server setup, domain controller configuration, user and group management, Group Policy configuration, Windows 11 domain joining, domain authentication, and account lockout investigation.

The lab also showed how Active Directory administration tasks can be verified using both graphical tools and command-line tools. By reviewing Event Viewer logs for failed logons and account lockouts, the project also included basic security investigation practice within a Windows domain environment.